PRACTICE TASK 3
LOPEZ, JANE RYZA S.
ZARATE, ARREL R.
BSCS - C303

1.Logical - IF

Note
If instead of marks, we had Grades A, B, C i.e. text values then the above formulas would have changed by
IF(B21= 'A',"Pass","Fail")

Exercise
Write a formula using IF. Salary for an AVP will be $2,000 and for an analyst will be $1,000

Emp ID	Designation	Salary	
E001	AVP	**$2,000**	=IF(B34="AVP",2000,1000)
E002	Analyst	**$1,000**	=IF(B35="AVP",2000,1000)
E003	Analyst	**$1,000**	=IF(B36="AVP",2000,1000)
E004	Analyst	**$1,000**	=IF(B37="AVP",2000,1000)

Logical - IF | Logical - AND | Logical - Nested IFs | Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE

2.Logical - AND

Example 2
Use IF with AND function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(AND(B26="PASS",C26="PASS"),"PASS","FAIL")
S002	PASS	FAIL	FAIL	=IF(AND(B27="PASS",C27="PASS"),"PASS","FAIL")
S003	FAIL	PASS	FAIL	=IF(AND(B28="PASS",C28="PASS"),"PASS","FAIL")
S004	PASS	PASS	PASS	=IF(AND(B29="PASS",C29="PASS"),"PASS","FAIL")

Exercise
Write a formula using AND with IF to find if an employee should get a "GOOD" increment or not. Sales made by the employee should be more than 50 and he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	
E001	AVP	**51**	**YES**	YES	=IF(AND(C36>50,D36="YES"),"YES","NO")
E002	Analyst	**100**	**NO**	NO	=IF(AND(C37>50,D37="YES"),"YES","NO")
E003	Analyst	**49**	**YES**	NO	=IF(AND(C38>50,D38="YES"),"YES","NO")
E004	Analyst	**75**	**YES**	YES	=IF(AND(C39>50,D39="YES"),"YES","NO")

Logical - IF | Logical - AND | Logical - Nested IFs | Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE

3.Logical - Nested IFs

Example

Use Nested IF to put the grade a student gets in an exam in Table B. Score Ranges for different grades are given in Table A.

TABLE A	
Score	**Grade**
> 70	A
41-70	B
< 40	C

TABLE B			
Student ID	**Marks**	**Grade**	**Formula**
S001	90	A	=IF(E26< 41,"C",IF(E26< 71,"B","A"))
S002	70	B	=IF(E27< 41,"C",IF(E27< 71,"B","A"))
S003	20	C	=IF(E28< 41,"C",IF(E28< 71,"B","A"))
S004	41	B	=IF(E29< 41,"C",IF(E29< 71,"B","A"))

Exercise

Write a formula using Nested IF conditions to check what level increment an employee should get and then convert the amount into dollars using Format

TABLE A	
Performance	**Increment**
A	$1,000
B	$500
C	$250

TABLE B				
Emp ID	**Designation**	**Performance**	**Result**	
E001	AVP	**A**	$ 1,000.00	=IF(F36="A",1000,IF(F36="B",500,IF(F36="C",250,"")))
E002	Analyst	**A**	$ 1,000.00	=IF(F37="A",1000,IF(F37="B",500,IF(F37="C",250,"")))
E003	Analyst	**B**	$ 500.00	=IF(F38="A",1000,IF(F38="B",500,IF(F38="C",250,"")))
E004	Analyst	**C**	$ 250.00	=IF(F39="A",1000,IF(F39="B",500,IF(F39="C",250,"")))

Logical - IF | Logical - AND | Logical - Nested IFs | Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE

4.Logical - OR

Example 2

Use IF with OR function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(OR(B28="FAIL",C28="FAIL"),"FAIL","PASS")
S002	PASS	FAIL	FAIL	=IF(OR(B29="FAIL",C29="FAIL"),"FAIL","PASS")
S003	FAIL	PASS	FAIL	=IF(OR(B30="FAIL",C30="FAIL"),"FAIL","PASS")
S004	PASS	PASS	PASS	=IF(OR(B31="FAIL",C31="FAIL"),"FAIL","PASS")

Exercise

Write a formula using OR with IF to find if an employee should get a "GOOD" increment or not. Either Sales made by the employee should be more than 50 or he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	
E001	AVP	**51**	**YES**	YES	=IF(OR(C38>50,D38="YES"),"YES","NO")
E002	Analyst	**49**	**NO**	NO	=IF(OR(C39>50,D39="YES"),"YES","NO")
E003	Analyst	**49**	**YES**	YES	=IF(OR(C40>50,D40="YES"),"YES","NO")
E004	Analyst	**75**	**YES**	YES	=IF(OR(C41>50,D41="YES"),"YES","NO")

Logical - IF | Logical - AND | Logical - Nested IFs | Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE

5.Logical - IFERROR

Note

In the above example, on dividing 1 by 0, we get an error, without using IFERROR we get #DIV/0! i.e. an error. So using IFERROR, we can give values we want. You can also enter text which can let the user know that there is some problem with the data.

Example 2

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	2	=A29*B29	2	=IFERROR(A29*B29,"DATA IS WRONG")
1	1	1	=A30*B30	1	=IFERROR(A30*B30,"DATA IS WRONG")
1	a	#VALUE!	=A31*B31	DATA IS WRONG	=IFERROR(A31*B31,"DATA IS WRONG")
5	a	#VALUE!	=A32*B32	DATA IS WRONG	=IFERROR(A32*B32,"DATA IS WRONG")

The red color highlight is for representation here - color cannot be changed using IFERROR

Exercise

Write a formula using IFERROR. Find the monthly salary of each employee using data in Column C and if an error appears, then change it to ERROR IN SALARY. Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary	
E001	AVP	£80,000	6667	=IFERROR(C39/12,"ERROR IN SALARY")
E002	Analyst	£50,000	4167	=IFERROR(C40/12,"ERROR IN SALARY")
E003	Analyst	abcd	ERROR IN SALARY	=IFERROR(C41/12,"ERROR IN SALARY")
E004	Analyst	£40,000	3333	=IFERROR(C42/12,"ERROR IN SALARY")

Logical - IF | Logical - AND | Logical - Nested IFs | Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE

6.Text - Exact

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Col A	Col B	Results	
TEST	Test	FALSE	=EXACT(A29,B29)
TEST	TEST	TRUE	=EXACT(A30,B30)
1234	1234	TRUE	=EXACT(A31,B31)
1.1	1.101	FALSE	=EXACT(A32,B32)

Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM

Accessibility: Investigate

7.Text - TEXT

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Value	Convert To	Results	
21/05/2017	21/05/2017	**21/05/2017**	=TEXT(A34, "dd/mm/yyyy")
42876	21/05/2017	**21/05/2017**	=TEXT(A35, "DD/MM/YYYY")
12:00	12:00	**12:00**	=TEXT(A36, "HH:MM")
12345.78	£12,345.78	**£12,345.78**	=TEXT(A37, "£#,##0.00")
12345.78	£12,346	**£12,346**	=TEXT(A38, "£#,##0")

Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE | Te

Accessibility: Investigate

8.Text-CONCATENATE

Exercise

Join the 3 strings together using CONCATENATE and & (ampersand) operator to get the results as mentioned in the result column

String1	String2	String3	Result	USE	Formula	
E001	AVP	800	**E001-AVP-800**	CONCATENATE	**E001-AVP-800**	=CONCATENATE(A35,"-",B35,"-",C35)
E002	Analyst	500	**E002Analyst500**	CONCATENATE	**E002Analyst500**	=CONCATENATE(A36,B36,C36)
E003	Analyst	100	**E003/Analyst-100**	CONCATENATE	**E003/Analyst-100**	=CONCATENATE(A37,"/",B37,"-",C37)
E004	Analyst	400	**E004:Analyst:400**	&	**E004:Analyst:400**	=A38&":"&B38&":"&C38
E005	AVP	1000	**E005 AVP 1000**	&	**E005 AVP 1000**	=A39&" "&B39&" "&C39
E006	Analyst	250	**E006 Analyst 250**	&	**E006 Analyst 250**	=A40&" "&B40&" "&C40
E007	Analyst	500	**E007;Analyst*500**	&	**E007;Analyst*500**	=A41&";"&B41&"*"&C41

Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE | Text - VALUE | Text - UPPER | Text -

Accessibility: Investigate

9. Text - TRIM

Exercise

Remove the extra spaces in the strings and then join the 3 strings together using CONCATENATE and & (ampersand) operator to get the results as mentioned in the result column. Do so in one formula itself

String1	String2	String3	Result	USE	Formula	
E001	AVP	800	**E001-AVP-800**	CONCATENATE	**E001**	=TRIM(A25)
E002	Analyst	500	**E002Analyst500**	CONCATENATE	**E002**	=TRIM(A26)
E003	Ana lyst	100	**E003/Analyst-100**	&	**Ana lyst**	=TRIM(B27)
E004	Analyst	400	**E004:Analyst:400**	&	**Analyst**	=TRIM(B28)

Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE | Text - VALUE | Text - UPPER | Text

Accessibility: Investigate

10. Text-SUBSTITUTE

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32 **Note**

33 If instances are not mentioned, then excel will replace all the occurrences as can be seen on Row 28. And the function is case-sensitive

34 which can be seen on Row 30

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36 **Exercise**

37 Use SUBSTITUTE function to make changes to some of the strings below to get the results mentioned in the 'Result' column

	String	Result	Formula	
38				
39	The dog is in the lawn	dog is in the lawn	**dog is in the lawn**	=SUBSTITUTE(A39,"The ","")
40	the dog is in the lawn	Dog is in lawn	**Dog is in lawn**	=SUBSTITUTE(SUBSTITUTE(A40,"the ",""),"dog","Dog")
41	Dog	Fog	**Fog**	=SUBSTITUTE(A41,"D","F")
42	Curious Cat	Furious Cat	**Furious Cat**	=SUBSTITUTE(A42,"C","F",1)

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Logical - AND | Logical - Nested IFs | Logical - OR | Logical - IFERROR | Text - Exact | Text - TEXT | Text - CONCATENATE | Text - TRIM | Text - SUBSTITUTE | Text - VALU